

0.1 曲线的挠率和 Frenet 公式

定义 0.1

设 $\beta(s)$ 和 $\gamma(s)$ 分别是正则曲线 C 的主法向量和次法向量, 其中 s 是曲线的弧长参数, 则 $\tau(s) = -\gamma'(s) \cdot \beta(s)$ 称为曲线 C 的**挠率**.

注

1. 根据定理??, 单位切向量 α 关于弧长参数 s 的导数的长度 $|\alpha'(s)|$ 反映了曲线的切线方向转动的快慢;
2. 由命题 0.1, 次法向量 γ 关于弧长参数 s 的导数的长度 $|\gamma'(s)|$ 反映了曲线的密切平面方向转动的快慢, 因而它刻画了曲线偏离平面曲线的程度, 反映了曲线扭曲的程度, 即曲线的“挠率”.

命题 0.1

设 $\alpha(s)$ 、 $\beta(s)$ 和 $\gamma(s)$ 分别是正则曲线 C 的单位切向量场、主法向量场和次法向量场, $\tau(s)$ 为曲线 C 的挠率, 其中 s 是曲线的弧长参数. 则

$$\gamma'(s) \text{ 与 } \beta(s) \text{ 共线, 且 } |\tau(s)| = |\gamma'(s)|.$$

进而

$$\gamma'(s) = -\tau(s) \cdot \beta(s), \quad \beta(s) = -\tau(s) \cdot \gamma'(s).$$

证明 因为 $\gamma(s)$ 是单位向量场, 所以 $\gamma^2(s) = 1$, 再求导可得 $\gamma(s)\gamma'(s) = 0$, 故 $\gamma'(s) \perp \gamma(s)$. 此外,

$$\gamma(s) = \alpha(s) \times \beta(s), \quad \alpha'(s) \parallel \beta(s),$$

所以

$$\gamma'(s) = \alpha'(s) \times \beta(s) + \alpha(s) \times \beta'(s) = \alpha(s) \times \beta'(s).$$

这说明 $\gamma'(s) \perp \alpha(s)$. 于是 $\gamma'(s)$ 必定是与 $\beta(s)$ 共线的, 不妨设

$$\gamma'(s) = -\tau\beta(s),$$

因此

$$\tau(s) = -\gamma'(s) \cdot \beta(s),$$

并且

$$|\tau(s)| = |\gamma'(s)|.$$

□

定理 0.1

设正则曲线 C 不是直线, 即曲率不是常数, 则它是平面曲线当且仅当它的挠率为零.

注 实际上, 若曲线 C 是直线, 则必然也是平面曲线.

证明 必要性: 若曲线 C 为平面曲线, 则曲线整体落在固定平面上, 该平面即为曲线的密切平面, 对应的次法向量 $\gamma(s)$ 为常向量. 而常向量关于弧长参数的导数满足 $\gamma'(s) \equiv 0$, 代入挠率定义式 $\tau(s) = -\gamma'(s) \cdot \beta(s)$, 直接可得 $\tau(s) \equiv 0$.

充分性: 设曲线 C 的参数方程是 $r = r(s)$, s 是弧长参数, 并且 $\kappa(s) \neq 0$, $\tau(s) \equiv 0$. 此时, 曲线有确定的 Frenet 标架 $\{r(s); \alpha(s), \beta(s), \gamma(s)\}$, 并且

$$\gamma'(s) = -\tau(s) \cdot \beta(s) \equiv 0,$$

因此 $\gamma(s) = \gamma_0$ 是常向量. 但是

$$0 = r'(s) \cdot \gamma(s) = r'(s) \cdot \gamma_0 = \frac{d}{ds}(r(s) \cdot \gamma_0),$$

所以 $\mathbf{r}(s) \cdot \gamma_0 = \mathbf{r}(s_0) \cdot \gamma_0$, 进而

$$(\mathbf{r}(s) - \mathbf{r}(s_0)) \cdot \gamma_0 = 0.$$

这说明曲线 $\mathbf{r} = \mathbf{r}(s)$ 落在经过点 $\mathbf{r}(s_0)$ 、以常向量 γ_0 为法向量的平面内.

□

定理 0.2 (Frenet 公式)

设正则曲线 C 的参数方程为 $\mathbf{r}(s)$, 其中 s 是弧长参数. 其 Frenet 标架为 $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$, 其中 $\boldsymbol{\alpha}(s) = \mathbf{r}'(s)$ 是单位切向量, $\boldsymbol{\beta}(s)$ 是主法向量, $\boldsymbol{\gamma}(s)$ 是副法向量. $\kappa(s)$ 和 $\tau(s)$ 分别为曲线 C 的曲率和挠率. 则曲线 C 的 Frenet 标架满足如下微分方程组:

$$\begin{cases} \mathbf{r}'(s) = \boldsymbol{\alpha}(s), \\ \boldsymbol{\alpha}'(s) = \kappa(s)\boldsymbol{\beta}(s), \\ \boldsymbol{\beta}'(s) = -\kappa(s)\boldsymbol{\alpha}(s) + \tau(s)\boldsymbol{\gamma}(s), \\ \boldsymbol{\gamma}'(s) = -\tau(s)\boldsymbol{\beta}(s). \end{cases} \quad (1)$$

上述公式称为 **Frenet**, 也可写作矩阵形式

$$\begin{pmatrix} \boldsymbol{\alpha}'(s) \\ \boldsymbol{\beta}'(s) \\ \boldsymbol{\gamma}'(s) \end{pmatrix} = \begin{pmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{pmatrix} \cdot \begin{pmatrix} \boldsymbol{\alpha}(s) \\ \boldsymbol{\beta}(s) \\ \boldsymbol{\gamma}(s) \end{pmatrix}.$$

♡

注 沿曲线定义的任意一个单位正交标架场的导数公式的系数矩阵都是反对称矩阵. Frenet 公式是曲线论中最重要、最基本的公式, 由法国数学家 Serret 和 Frenet 分别在 1851 年和 1852 年发表 (他们给出的是切线、主法线、次法线的方向余弦的导数公式).

证明 由已知条件, $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$ 是空间 E^3 的一个单位正交标架, 因此 $\boldsymbol{\beta}'(s)$ 可表示为 $\boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)$ 的线性组合. 设

$$\boldsymbol{\beta}'(s) = a\boldsymbol{\alpha}(s) + b\boldsymbol{\beta}(s) + c\boldsymbol{\gamma}(s). \quad (2)$$

分别与 $\boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)$ 作点乘, 再结合 $\boldsymbol{\beta}(s) \cdot \boldsymbol{\alpha}(s) = \boldsymbol{\beta}(s) \cdot \boldsymbol{\gamma}(s) = 0$ 求导后仍为 0, 以及单位正交性可得

$$a = \boldsymbol{\beta}'(s) \cdot \boldsymbol{\alpha}(s) = -\boldsymbol{\beta}(s) \cdot \boldsymbol{\alpha}'(s) = -\kappa(s),$$

$$b = \boldsymbol{\beta}'(s) \cdot \boldsymbol{\beta}(s) = 0,$$

$$c = \boldsymbol{\beta}'(s) \cdot \boldsymbol{\gamma}(s) = -\boldsymbol{\beta}(s) \cdot \boldsymbol{\gamma}'(s) = \tau(s).$$

代入 (2) 即得 (1) 式中的第三个公式. 又由定义 ?? 和 命题 0.1 可知, $\mathbf{r}'(s) = \boldsymbol{\alpha}(s)$ 、 $\boldsymbol{\alpha}'(s) = \kappa(s)\boldsymbol{\beta}(s)$ 和 $\boldsymbol{\gamma}'(s) = -\tau(s)\boldsymbol{\beta}(s)$, 这样就得到完整的 Frenet 公式. □

定理 0.3

(1) 设正则曲线 $\mathbf{r} = \mathbf{r}(s)$ 的曲率 $\kappa(s)$ 和挠率 $\tau(s)$ 都不为零, s 是弧长参数. 如果该曲线落在一个球面上, 则它的曲率和挠率必满足关系式

$$\left(\frac{1}{\kappa(s)}\right)^2 + \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)}\right)\right)^2 = C,$$

其中 C 为一个常数. 并且

$$\frac{\tau(s)}{\kappa(s)} + \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)}\right)\right) = 0.$$

(2) 设正则曲线 $\mathbf{r} = \mathbf{r}(s)$ 的曲率 $\kappa(s)$ 和挠率 $\tau(s)$ 都不为零, s 是弧长参数. 并且 $\kappa(s)$ 不是常数, 则该曲线落在一个球面上的充要条件是它的曲率和挠率满足关系式

$$\left(\frac{1}{\kappa(s)}\right)^2 + \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)}\right)\right)^2 = C,$$

其中 C 为一个常数.

证明

- (1) 设曲线的 Frenet 标架为 $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$, 其中 $\boldsymbol{\alpha}(s) = \mathbf{r}'(s)$ 是单位切向量, $\boldsymbol{\beta}(s)$ 是主法向量, $\boldsymbol{\gamma}(s)$ 是副法向量. 假定曲线 $\mathbf{r} = \mathbf{r}(s)$ 落在一个球面上, 该球面的球心是 $\mathbf{r}_0$, 半径是 a , 则有关系式

$$(\mathbf{r}(s) - \mathbf{r}_0)^2 = a^2. \quad (3)$$

将上式两边对于 s 求导, 得到

$$\boldsymbol{\alpha}(s) \cdot (\mathbf{r}(s) - \mathbf{r}_0) = 0,$$

故 $(\mathbf{r}(s) - \mathbf{r}_0)$ 落在 $\boldsymbol{\alpha}(s)$ 的二维正交补空间 $\text{span}\{\boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$ 里. 不妨设

$$\mathbf{r}(s) - \mathbf{r}_0 = \lambda(s)\boldsymbol{\beta}(s) + \mu(s)\boldsymbol{\gamma}(s). \quad (4)$$

将上式对于 s 求导并且利用 Frenet 公式得到

$$\boldsymbol{\alpha}(s) = -\lambda(s)\kappa(s)\boldsymbol{\alpha}(s) + (\lambda'(s) - \mu(s)\tau(s))\boldsymbol{\beta}(s) + (\lambda(s)\tau(s) + \mu'(s))\boldsymbol{\gamma}(s).$$

因此比较等式两边的系数得到

$$\lambda(s)\kappa(s) = -1, \quad \lambda'(s) = \mu(s)\tau(s), \quad \mu'(s) = -\lambda(s)\tau(s), \quad (5)$$

于是

$$\lambda(s) = -\frac{1}{\kappa(s)}, \quad \mu(s) = \frac{\lambda'(s)}{\tau(s)} = -\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right). \quad (6)$$

将(6)式代入(4)式得到

$$\mathbf{r}(s) - \mathbf{r}_0 = -\frac{1}{\kappa(s)}\boldsymbol{\beta}(s) - \frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \boldsymbol{\gamma}(s).$$

因此根据关系式(3)和单位正交性得到

$$\left(\frac{1}{\kappa(s)} \right)^2 + \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right)^2 = a^2.$$

将(6)式代入(5)的第三式便得到

$$\frac{\tau(s)}{\kappa(s)} + \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) = 0.$$

- (2) 必要性由(1)立得. 下面证明充分性. 设曲线的 Frenet 标架为 $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$, 其中 $\boldsymbol{\alpha}(s) = \mathbf{r}'(s)$ 是单位切向量, $\boldsymbol{\beta}(s)$ 是主法向量, $\boldsymbol{\gamma}(s)$ 是副法向量. 记 $f(s) = \mathbf{r}(s) + \frac{1}{\kappa(s)}\boldsymbol{\beta}(s) + \frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \boldsymbol{\gamma}(s)$, 则由条件和 Frenet 公式可知

$$\begin{aligned} & \frac{d}{ds} \left(\left(\frac{1}{\kappa(s)} \right)^2 + \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right)^2 \right) = 0 \\ \iff & \frac{2}{\kappa(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) + \frac{2}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \cdot \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) = 0 \\ \iff & \frac{1}{\kappa(s)} + \frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) = 0 \\ \iff & \frac{\tau(s)}{\kappa(s)} + \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) = 0. \end{aligned}$$

于是再利用 Frenet 公式可得

$$\begin{aligned} f'(s) &= \mathbf{r}'(s) + \boldsymbol{\beta}(s) \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) + \frac{1}{\kappa(s)} \boldsymbol{\beta}'(s) + \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) \boldsymbol{\gamma}(s) + \frac{\boldsymbol{\gamma}'(s)}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \\ &= \boldsymbol{\alpha}(s) + \boldsymbol{\beta}(s) \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) + \frac{1}{\kappa(s)} [-\kappa(s)\boldsymbol{\alpha}(s) + \tau(s)\boldsymbol{\gamma}(s)] + \boldsymbol{\gamma}(s) \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) - \boldsymbol{\beta}(s) \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \\ &= \boldsymbol{\gamma}(s) \left[\frac{\tau(s)}{\kappa(s)} + \frac{d}{ds} \left(\frac{1}{\tau(s)} \frac{d}{ds} \left(\frac{1}{\kappa(s)} \right) \right) \right] = 0. \end{aligned}$$

故存在 $\mathbf{r}_0 \in \mathbb{R}^3$, 使得

$$\mathbf{r}_0 = f(s) = \mathbf{r}(s) + \frac{1}{\kappa(s)}\boldsymbol{\beta}(s) + \frac{1}{\tau(s)}\frac{d}{ds}\left(\frac{1}{\kappa(s)}\right)\boldsymbol{\gamma}(s).$$

再利用 $\boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)$ 的单位正交性可得

$$|\mathbf{r}(s) - \mathbf{r}_0|^2 = \left| \frac{1}{\kappa(s)}\boldsymbol{\beta}(s) + \frac{1}{\tau(s)}\frac{d}{ds}\left(\frac{1}{\kappa(s)}\right)\boldsymbol{\gamma}(s) \right|^2 = \left(\frac{1}{\kappa(s)}\right)^2 + \left(\frac{1}{\tau(s)}\frac{d}{ds}\left(\frac{1}{\kappa(s)}\right)\right)^2 = C.$$

因此该曲线落在一个球面上.

□

定理 0.4 (曲线挠率的计算公式)

1. 设曲线 $\mathbf{r} = \mathbf{r}(s)$ 是正则参数曲线, 其中 s 为弧长参数. 其曲率 $\kappa(s) \neq 0$, 则其挠率为

$$\tau(s) = \frac{(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s))}{|\mathbf{r}''(s)|^2} = \frac{(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s))}{\kappa^2(s)}. \quad (7)$$

其中 $(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s))$ 表示三个向量的混合积.

2. 设曲线 $\mathbf{r} = \mathbf{r}(t)$ 是正则参数曲线, 其中 t 为任意参数. 其曲率 $\kappa(t) \neq 0$, 则其挠率为

$$\tau(t) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2}. \quad (8)$$

其中 $(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))$ 表示三个向量的混合积.

♡

证明

1. 设曲线的 Frenet 标架为 $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$, 其中 $\boldsymbol{\alpha}(s) = \mathbf{r}'(s)$ 是单位切向量, $\boldsymbol{\beta}(s)$ 是主法向量, $\boldsymbol{\gamma}(s)$ 是副法向量. 由曲线论基本公式知

$$\kappa(s) = |\mathbf{r}''(s)| \neq 0, \quad \boldsymbol{\beta}(s) = \frac{\mathbf{r}''(s)}{|\mathbf{r}''(s)|}, \quad \boldsymbol{\gamma}(s) = \boldsymbol{\alpha}(s) \times \boldsymbol{\beta}(s) = \frac{\mathbf{r}'(s) \times \mathbf{r}''(s)}{|\mathbf{r}''(s)|}.$$

对 $\boldsymbol{\gamma}(s)$ 求导得

$$\boldsymbol{\gamma}'(s) = \frac{d}{ds} \left(\frac{\mathbf{r}'(s) \times \mathbf{r}''(s)}{|\mathbf{r}''(s)|} \right) = \frac{\mathbf{r}'(s) \times \mathbf{r}'''(s)}{|\mathbf{r}''(s)|} + (\mathbf{r}'(s) \times \mathbf{r}''(s)) \frac{d}{ds} \left(\frac{1}{|\mathbf{r}''(s)|} \right).$$

两边与 $\boldsymbol{\beta}(s) = \frac{\mathbf{r}''(s)}{|\mathbf{r}''(s)|}$ 作点乘. 又由挠率定义知 $\boldsymbol{\gamma}'(s) \cdot \boldsymbol{\beta}(s) = -\tau(s)$, 因此

$$-\tau(s) = \frac{(\mathbf{r}'(s) \times \mathbf{r}'''(s)) \cdot \mathbf{r}''(s)}{|\mathbf{r}''(s)|^2} = -\frac{(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s))}{|\mathbf{r}''(s)|^2}.$$

上式两边同乘 -1 即得(7)式.

2. 设曲线的 Frenet 标架为 $\{\mathbf{r}(t); \boldsymbol{\alpha}(t), \boldsymbol{\beta}(t), \boldsymbol{\gamma}(t)\}$, 其中 $\boldsymbol{\alpha}(t) = \mathbf{r}'(t)$ 是单位切向量, $\boldsymbol{\beta}(t)$ 是主法向量, $\boldsymbol{\gamma}(t)$ 是副法向量. 由弧长微分 $\frac{ds}{dt} = |\mathbf{r}'(t)|$ 及曲线论基本公式可得

$$\begin{aligned} \boldsymbol{\alpha}(t) &= \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}, \quad \boldsymbol{\gamma}(t) = \frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}, \\ \boldsymbol{\beta}(t) &= \boldsymbol{\gamma}(t) \times \boldsymbol{\alpha}(t) = \frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \mathbf{r}''(t) - \frac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{|\mathbf{r}'(t)| \cdot |\mathbf{r}'(t) \times \mathbf{r}''(t)|} \mathbf{r}'(t). \end{aligned} \quad (9)$$

将 $\boldsymbol{\gamma}(t)$ 对 t 求导得

$$\frac{d\boldsymbol{\gamma}}{dt} = \frac{d}{dt} \left(\frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \right) = \frac{\mathbf{r}'(t) \times \mathbf{r}'''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} + \frac{d}{dt} \left(\frac{1}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \right) (\mathbf{r}'(t) \times \mathbf{r}''(t)).$$

利用 Frenet 公式中的 $\frac{d\boldsymbol{\gamma}}{ds} = -\tau(s)\boldsymbol{\beta}(s)$ 以及链式法则可知

$$\frac{d\boldsymbol{\gamma}}{dt} = \frac{d\boldsymbol{\gamma}}{ds} \frac{ds}{dt} = -\tau(t) \frac{ds}{dt} \boldsymbol{\beta}(t).$$

进而

$$-\tau(t) \frac{ds}{dt} \boldsymbol{\beta}(t) = \frac{d}{dt} \left(\frac{\mathbf{r}'(t) \times \mathbf{r}''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \right) = \frac{\mathbf{r}'(t) \times \mathbf{r}'''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} + \frac{d}{dt} \left(\frac{1}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \right) (\mathbf{r}'(t) \times \mathbf{r}''(t)).$$

然后两边与 $\beta(t)$ 作点乘. 又注意到 $\frac{ds}{dt} = |\mathbf{r}'(t)|$, 再结合(9)式可得

$$\begin{aligned} -\tau(t)|\mathbf{r}'(t)| &= -\tau(t)\frac{ds}{dt} = \frac{\mathbf{r}'(t) \times \mathbf{r}'''(t)}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \cdot \frac{|\mathbf{r}'(t)|}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|} \mathbf{r}''(t) \\ &= \frac{|\mathbf{r}'(t)|(\mathbf{r}''(t), \mathbf{r}'(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = -\frac{|\mathbf{r}'(t)|(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2}. \end{aligned}$$

上式两边同乘 -1 即得(8)式. □

定理 0.5

曲线 $\mathbf{r} = \mathbf{r}(t)$ 是一条平面曲线的充分必要条件是

$$(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) \equiv 0.$$

证明 由定理 0.1 和曲线挠率的计算公式立得. □

命题 0.2

设 E^3 中的一条正则曲线 C 的参数方程为 $\mathbf{r} = \mathbf{r}(s)$, 其中 s 是弧长参数. 再设 σ 是 E^3 的一个刚体运动, 曲线 C 点 $\mathbf{r}(s)$ 处的曲率为 $\kappa(s)$, 挠率为 $\tau(s)$. 记曲线 $\sigma(\mathbf{r}(s)) = \sigma(\mathbf{r}(s))$ 的曲率为 $\kappa_\sigma(s)$, 挠率为 $\tau_\sigma(s)$, 则

$$\kappa_\sigma(s) = \kappa(s), \quad \tau_\sigma(s) = \tau(s), \quad \forall s.$$

证明 由定理??知存在 $\mathbf{a} \in E^3, A \in SO(3)$, 使得

$$\sigma(\mathbf{x}) = \mathbf{a} + A\mathbf{x}, \quad \forall \mathbf{x} \in E^3.$$

于是

$$\sigma(\mathbf{r}(s)) = \mathbf{a} + A\mathbf{r}(s), \quad \forall s.$$

从而

$$\frac{d\sigma(\mathbf{r}(s))}{ds} = A\mathbf{r}'(s), \quad \frac{d^2\sigma(\mathbf{r}(s))}{ds^2} = A\mathbf{r}''(s), \quad \frac{d^3\sigma(\mathbf{r}(s))}{ds^3} = A\mathbf{r}'''(s).$$

因此

$$\begin{aligned} (A\mathbf{r}'(s), A\mathbf{r}''(s), A\mathbf{r}'''(s)) &= \det(A)(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s)) \\ &= (\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s)) \quad (\det(A) = 1). \end{aligned}$$

由曲线论基本公式和曲线挠率的计算公式的计算公式可得

$$\begin{aligned} \kappa_\sigma(s) &= \left| \frac{d^2\sigma(\mathbf{r}(s))}{ds^2} \right| = |A\mathbf{r}''(s)| = |\mathbf{r}''(s)| = \kappa(s), \\ \tau_\sigma(s) &= \frac{(\frac{d\sigma(\mathbf{r}(s))}{ds}, \frac{d^2\sigma(\mathbf{r}(s))}{ds^2}, \frac{d^3\sigma(\mathbf{r}(s))}{ds^3})}{\left| \frac{d^2\sigma(\mathbf{r}(s))}{ds^2} \right|^2} = \frac{(\mathbf{r}'(s), \mathbf{r}''(s), \mathbf{r}'''(s))}{|\mathbf{r}''(s)|^2} = \tau(s). \end{aligned}$$

□

例题 0.1 求下列曲线的挠率:

- (1) $\mathbf{r}(t) = (at, a\sqrt{2}\ln t, \frac{a}{t}), a > 0;$
- (2) $\mathbf{r}(t) = (3t - t^3, 3t^2, 3t + t^3);$
- (3) $\mathbf{r}(t) = (a(t - \sin t), a(1 - \cos t), bt), a > 0;$
- (4) $\mathbf{r}(t) = (\cos^3 t, \sin^3 t, \cos 2t).$

解

(1) 由题设可得

$$\mathbf{r}'(t) = \left(a, \frac{a\sqrt{2}}{t}, -\frac{a}{t^2}\right), \quad \mathbf{r}''(t) = \left(0, -\frac{a\sqrt{2}}{t^2}, \frac{2a}{t^3}\right), \quad \mathbf{r}'''(t) = \left(0, \frac{2\sqrt{2}a}{t^3}, -\frac{6a}{t^4}\right),$$

进而

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \left(\frac{a^2\sqrt{2}}{t^4}, -\frac{2a^2}{t^3}, -\frac{a^2\sqrt{2}}{t^2}\right), \quad |\mathbf{r}'(t) \times \mathbf{r}''(t)| = \frac{a^2\sqrt{2}}{t^2} \left(1 + \frac{1}{t^2}\right), \quad (\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) = \frac{2\sqrt{2}a^3}{t^6}.$$

故由曲线挠率的计算公式可知所求曲线的挠率为

$$\tau(t) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = \frac{\frac{2\sqrt{2}a^3}{t^6}}{\frac{2a^4}{t^4} \left(1 + \frac{1}{t^2}\right)^2} = \frac{\sqrt{2}t^2}{a(1+t^2)^2}.$$

(2) 由题设可得

$$\mathbf{r}'(t) = (3-3t^2, 6t, 3+3t^2), \quad \mathbf{r}''(t) = (-6t, 6, 6t), \quad \mathbf{r}'''(t) = (-6, 0, 6),$$

进而

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = (-18t^2+18, -36t, 18t^2+18), \quad |\mathbf{r}'(t) \times \mathbf{r}''(t)| = 18\sqrt{2}(1+t^2), \quad (\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) = 216,$$

故由曲线挠率的计算公式可知所求曲线的挠率为

$$\tau(t) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = \frac{216}{2 \cdot 18^2(1+t^2)^2} = \frac{1}{3(1+t^2)^2}.$$

(3) 由题设可得

$$\mathbf{r}'(t) = (a-a\cos t, a\sin t, b), \quad \mathbf{r}''(t) = (a\sin t, a\cos t, 0), \quad \mathbf{r}'''(t) = (a\cos t, -a\sin t, 0),$$

进而

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = (-ab\cos t, ab\sin t, a^2(\cos t-1)), \quad |\mathbf{r}'(t) \times \mathbf{r}''(t)| = a\sqrt{b^2+a^2(\cos t-1)^2},$$

$$(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) = -a^2b,$$

故由曲线挠率的计算公式可知所求曲线的挠率为

$$\tau(t) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = \frac{-a^2b}{a^2b^2+a^4(\cos t-1)^2} = -\frac{b}{b^2+a^2(\cos t-1)^2}.$$

(4) 由题设可得

$$\mathbf{r}'(t) = (-3\sin t \cos^2 t, 3\sin^2 t \cos t, -2\sin 2t), \quad \mathbf{r}''(t) = (3\cos t(2\sin^2 t - \cos^2 t), 3\sin t(2\cos^2 t - \sin^2 t), -4\cos 2t),$$

$$\mathbf{r}'''(t) = (-6\sin^3 t + 21\sin t \cos^2 t, 6\cos^3 t - 21\sin^2 t \cos t, 8\sin 2t),$$

进而

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = (12\sin^2 t \cos^3 t, -12\sin^3 t \cos^2 t, -9\sin^2 t \cos^2 t), \quad |\mathbf{r}'(t) \times \mathbf{r}''(t)| = 15\sin^2 t \cos^2 t,$$

$$(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) = 36\sin^3 t \cos^3 t,$$

故由曲线挠率的计算公式可知所求曲线的挠率为

$$\tau(t) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = \frac{36\sin^3 t \cos^3 t}{15^2 \sin^4 t \cos^4 t} = -\frac{8}{25} \csc 2t.$$

□

例题 0.2 求曲线

$$\begin{cases} x + \sinh x = y + \sin y, \\ z + e^z = (x+1) + \ln(x+1) \end{cases}$$

在 $(0, 0, 0)$ 处的挠率.

解 设曲线的参数方程是

$$\mathbf{r}(s) = (x(s), y(s), z(s)),$$

其中 s 是曲线的弧长参数, 并且 $s = 0$ 对应于点 $(0, 0, 0)$, 即 $\mathbf{r}(0) = (0, 0, 0)$. 再结合定理??知, 函数 $x(s), y(s), z(s)$ 满足

$$\begin{cases} x(s) + \sinh x(s) = y(s) + \sin y(s), \\ z(s) + e^{z(s)} = (x(s) + 1) + \ln(x(s) + 1), \\ (x'(s))^2 + (y'(s))^2 + (z'(s))^2 = 1. \end{cases}$$

对上式中的前两式关于 s 求导得

$$\begin{cases} x'(s) + x'(s) \cosh x(s) = y'(s) + y'(s) \cos y(s), \\ z'(s) + z'(s) e^{z(s)} = x'(s) + \frac{x'(s)}{x(s) + 1}, \\ (x'(s))^2 + (y'(s))^2 + (z'(s))^2 = 1. \end{cases} \quad (10)$$

令 $s = 0$ 得

$$x'(0) = y'(0) = z'(0) = \frac{1}{\sqrt{3}}.$$

进而

$$\alpha(0) = \mathbf{r}'(0) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right).$$

对方程组(10)中三式都关于 s 求导得

$$\begin{cases} x''(s) + x''(s) \cosh x(s) + (x'(s))^2 \sinh x(s) = y''(s) + y''(s) \cos y(s) - (y'(s))^2 \sin y(s), \\ z''(s) + z''(s) e^{z(s)} + (z'(s))^2 e^{z(s)} = x''(s) + \frac{x''(s)}{x(s) + 1} - \frac{(x'(s))^2}{(x(s) + 1)^2}, \\ x'(s)x''(s) + y'(s)y''(s) + z'(s)z''(s) = 0. \end{cases} \quad (11)$$

再令 $s = 0$ 得

$$x''(0) = y''(0) = \frac{1}{9}, \quad z''(0) = -\frac{2}{9}.$$

从而

$$\mathbf{r}''(0) = \left(\frac{1}{9}, \frac{1}{9}, -\frac{2}{9} \right).$$

再对方程组(11)中三式都关于 s 求导得

$$\begin{aligned} & x'''(s)(1 + \cosh x(s)) + 3x'(s)x''(s)\sinh x(s) + (x'(s))^3 \cosh x(s) \\ &= y'''(s)(1 + \cos y(s)) - 3y'(s)y''(s)\sin y(s) - (y'(s))^3 \cos y(s), \\ & z'''(s)(1 + e^{z(s)}) + 3z'(s)z''(s)e^{z(s)} + (z'(s))^3 e^{z(s)} \\ &= x'''(s)\left(1 + \frac{1}{x(s) + 1}\right) - \frac{3x'(s)x''(s)}{(x(s) + 1)^2} + \frac{2(x'(s))^3}{(x(s) + 1)^3}, \\ & (x''(s))^2 + x'(s)x'''(s) + (y''(s))^2 + y'(s)y'''(s) + (z''(s))^2 + z'(s)z'''(s) = 0. \end{aligned}$$

再令 $s = 0$ 得

$$x'''(0) = -\frac{8\sqrt{3}}{81}, \quad y'''(0) = \frac{\sqrt{3}}{81}, \quad z'''(0) = \frac{\sqrt{3}}{81}.$$

即 $\mathbf{r}'''(0) = \left(-\frac{8\sqrt{3}}{81}, \frac{\sqrt{3}}{81}, \frac{\sqrt{3}}{81} \right)$. 综上所述可知

$$\mathbf{r}'(0) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \quad \mathbf{r}''(0) = \left(\frac{1}{9}, \frac{1}{9}, -\frac{2}{9} \right), \quad \mathbf{r}'''(0) = \left(-\frac{8\sqrt{3}}{81}, \frac{\sqrt{3}}{81}, \frac{\sqrt{3}}{81} \right).$$

于是

$$|\mathbf{r}''(0)| = \sqrt{\frac{6}{81}}, \quad (\mathbf{r}'(0), \mathbf{r}''(0), \mathbf{r}'''(0)) = \frac{1}{27}.$$

故由曲线挠率的计算公式可知所求曲线的挠率为

$$\tau(0) = \frac{(\mathbf{r}'(0), \mathbf{r}''(0), \mathbf{r}'''(0))}{|\mathbf{r}''(0)|^2} = \frac{\frac{1}{27}}{\frac{6}{81}} = \frac{1}{2}.$$

□

例题 0.3 假定曲线 $\mathbf{r} = \mathbf{r}(s)$ 的挠率 τ 是一个非零的常数, s 是弧长参数, 求曲线

$$\tilde{\mathbf{r}}(s) = \frac{1}{\tau} \boldsymbol{\beta}(s) - \int \boldsymbol{\gamma}(s) ds$$

的曲率和挠率.

注 这里 s 是曲线 $\mathbf{r} = \mathbf{r}(s)$ 的弧长参数, 不一定是 $\tilde{\mathbf{r}}(s)$ 的弧长参数.

解 由 Frenet 公式知

$$\begin{cases} \boldsymbol{\alpha}'(s) = \kappa(s) \boldsymbol{\beta}(s), \\ \boldsymbol{\beta}'(s) = -\kappa(s) \boldsymbol{\alpha}(s) + \tau \boldsymbol{\gamma}(s), \\ \boldsymbol{\gamma}'(s) = -\tau \boldsymbol{\beta}(s). \end{cases}$$

$$\text{从而 } \tilde{\mathbf{r}}'(s) = \frac{1}{\tau} \boldsymbol{\beta}'(s) - \boldsymbol{\gamma}(s) = -\frac{1}{\tau} \kappa(s) \boldsymbol{\alpha}(s), \quad \tilde{\mathbf{r}}''(s) = -\frac{1}{\tau} \kappa'(s) \boldsymbol{\alpha}(s) - \frac{1}{\tau} \kappa(s) \boldsymbol{\alpha}'(s) = -\frac{1}{\tau} [\kappa'(s) \boldsymbol{\alpha}(s) + \kappa^2(s) \boldsymbol{\beta}(s)],$$

$$\begin{aligned} \tilde{\mathbf{r}}'''(s) &= -\frac{1}{\tau} [\kappa''(s) \boldsymbol{\alpha}(s) + \kappa'(s) \boldsymbol{\alpha}'(s) + 2\kappa(s) \kappa'(s) \boldsymbol{\beta}(s) + \kappa^2(s) \boldsymbol{\beta}'(s)] \\ &= -\frac{1}{\tau} [(\kappa''(s) - \kappa^3(s)) \boldsymbol{\alpha}(s) + 3\kappa(s) \kappa'(s) \boldsymbol{\beta}(s) + \tau \kappa^2(s) \boldsymbol{\gamma}(s)]. \end{aligned}$$

由曲线挠率的计算公式, 再结合 $|\boldsymbol{\alpha}(s)| = 1$, 可得

$$\tilde{\kappa}(s) = \frac{|\tilde{\mathbf{r}}'(s) \times \tilde{\mathbf{r}}''(s)|}{|\tilde{\mathbf{r}}'(s)|^3} = \frac{\frac{1}{\tau^2} |\kappa(s) \boldsymbol{\alpha}(s) \times \kappa^2(s) \boldsymbol{\beta}(s)|}{\left| \frac{1}{\tau} \kappa(s) \boldsymbol{\alpha}(s) \right|^3} = \frac{\tau \kappa^3(s) |\boldsymbol{\alpha}(s) \times \boldsymbol{\beta}(s)|}{\kappa^3(s)} = \tau.$$

再结合 $|\boldsymbol{\gamma}(s)| = 1$, 可得

$$\begin{aligned} (\tilde{\mathbf{r}}'(s), \tilde{\mathbf{r}}''(s), \tilde{\mathbf{r}}'''(s)) &= (\tilde{\mathbf{r}}'(s) \times \tilde{\mathbf{r}}''(s)) \cdot \tilde{\mathbf{r}}'''(s) \\ &= -\frac{1}{\tau^3} \kappa^3(s) (\boldsymbol{\alpha}(s) \times \boldsymbol{\beta}(s)) \cdot \tau \kappa^2(s) \boldsymbol{\gamma}(s) \\ &= -\frac{1}{\tau^2} \kappa^5(s) \boldsymbol{\gamma}^2(s) = -\frac{1}{\tau^2} \kappa^5(s). \end{aligned}$$

故由曲线论基本公式可得

$$\tilde{\tau}(s) = \frac{(\tilde{\mathbf{r}}'(s), \tilde{\mathbf{r}}''(s), \tilde{\mathbf{r}}'''(s))}{|\tilde{\mathbf{r}}'(s) \times \tilde{\mathbf{r}}''(s)|^2} = \frac{-\frac{1}{\tau^2} \kappa^5(s)}{\frac{1}{\tau^4} \kappa^6(s)} = -\frac{\tau^2}{\kappa(s)}.$$

□

例题 0.4 证明: 若一条正则曲线的曲率处处不是零, 并且它在各点的密切平面都经过一个固定点, 则它必定落在一个平面上.

证明 设这条正则曲线 C 的参数方程为 $\mathbf{r} = \mathbf{r}(s)$, 其中 s 为弧长参数. 曲线 C 在各点的 Frenet 标架为 $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$, 其中 $\boldsymbol{\alpha}(s)$ 是单位切向量, $\boldsymbol{\beta}(s)$ 是主法向量, $\boldsymbol{\gamma}(s)$ 是副法向量. 则曲线 C 在各点的密切平面方程为

$$(\mathbf{X} - \mathbf{r}(s)) \cdot \boldsymbol{\gamma}(s) = 0.$$

不妨设曲线 C 的各点的密切平面都过原点, 则显然当 $s = 0$ 时, 有 $\tau(0) = 0$. 现在对任意的 $s \neq 0$, 都有

$$\mathbf{r}(s) \cdot \boldsymbol{\gamma}(s) = 0.$$

对上式关于 s 求导, 结合 Frenet 公式得

$$\mathbf{r}'(s) \cdot \boldsymbol{\gamma}(s) + \mathbf{r}(s) \cdot \boldsymbol{\gamma}'(s) = \boldsymbol{\alpha}(s) \cdot \boldsymbol{\gamma}(s) - \tau(s) \mathbf{r}(s) \cdot \boldsymbol{\beta}(s) = 0 \implies \tau(s) \mathbf{r}(s) \cdot \boldsymbol{\beta}(s) = 0.$$

若 $\tau(s) \neq 0$, 则 $\mathbf{r}(s) \cdot \boldsymbol{\beta}(s) = 0$. 再关于 s 求导, 结合 Frenet 公式得

$$\mathbf{r}'(s) \cdot \boldsymbol{\beta}(s) + \mathbf{r}(s) \cdot \boldsymbol{\beta}'(s) = \boldsymbol{\alpha}(s) \cdot \boldsymbol{\beta}(s) + \mathbf{r}(s) \cdot [-\kappa(s) \boldsymbol{\alpha}(s) + \tau(s) \boldsymbol{\gamma}(s)] = -\kappa(s) \mathbf{r}(s) \cdot \boldsymbol{\alpha}(s) = 0.$$

由 $\kappa(s) \neq 0$ 知 $\mathbf{r}(s) \cdot \boldsymbol{\alpha}(s) = 0$. 故 $\mathbf{r}(s) \in \text{span}^\perp \{\boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\} = \mathbb{R}^3$, 因此 $\mathbf{r}(s) = \mathbf{0}$, 矛盾! 故 $\tau(s) = 0$, 再结合 s 的任意性知, $\tau(s) \equiv 0$. 由定理 0.1 知, 曲线 C 是平面曲线. □

例题 0.5 设 $\mathbf{r} = \mathbf{r}(s)$ 是以 s 为弧长参数的正则参数曲线, 它的 Frenet 标架是 $\{\mathbf{r}(s); \boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$, 试求沿曲线定义的向量场 $\boldsymbol{\rho}(s)$, 使得

$$\boldsymbol{\alpha}'(s) = \boldsymbol{\rho}(s) \times \boldsymbol{\alpha}(s), \quad \boldsymbol{\beta}'(s) = \boldsymbol{\rho}(s) \times \boldsymbol{\beta}(s), \quad \boldsymbol{\gamma}'(s) = \boldsymbol{\rho}(s) \times \boldsymbol{\gamma}(s).$$

解 因为 $\{\boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)\}$ 是 $\mathbb{R}^3$ 的一组标准正交基, 所以可设

$$\boldsymbol{\rho}(s) = a(s)\boldsymbol{\alpha}(s) + b(s)\boldsymbol{\beta}(s) + c(s)\boldsymbol{\gamma}(s).$$

从而再结合 Frenet 公式可得

$$\kappa(s)\boldsymbol{\beta}(s) = \boldsymbol{\alpha}'(s) = \boldsymbol{\rho}(s) \times \boldsymbol{\alpha}(s) = -b(s)\boldsymbol{\gamma}(s) + c(s)\boldsymbol{\beta}(s),$$

$$-\kappa(s)\boldsymbol{\alpha}(s) + \tau(s)\boldsymbol{\gamma}(s) = \boldsymbol{\beta}'(s) = \boldsymbol{\rho}(s) \times \boldsymbol{\beta}(s) = -c(s)\boldsymbol{\alpha}(s) + a(s)\boldsymbol{\gamma}(s),$$

$$-\tau(s)\boldsymbol{\beta}(s) = \boldsymbol{\gamma}'(s) = \boldsymbol{\rho}(s) \times \boldsymbol{\gamma}(s) = b(s)\boldsymbol{\alpha}(s) - a(s)\boldsymbol{\beta}(s).$$

再结合 $\boldsymbol{\alpha}(s), \boldsymbol{\beta}(s), \boldsymbol{\gamma}(s)$ 线性无关可知

$$a(s) = \tau(s), \quad b(s) = 0, \quad c(s) = \kappa(s).$$

故

$$\boldsymbol{\rho}(s) = \tau(s)\boldsymbol{\alpha}(s) + \kappa(s)\boldsymbol{\gamma}(s).$$
□

例题 0.6 求曲线

$$\mathbf{r}(t) = \left(\frac{(1+t)^{\frac{3}{2}}}{3}, \frac{(1-t)^{\frac{3}{2}}}{3}, \frac{t}{\sqrt{2}} \right), \quad -1 < t < 1$$

的曲率、挠率和 Frenet 标架场.

解 由条件可得

$$\begin{aligned} \mathbf{r}'(t) &= \left(\frac{1}{2}\sqrt{1+t}, -\frac{1}{2}\sqrt{1-t}, \frac{1}{\sqrt{2}} \right), & \mathbf{r}''(t) &= \left(\frac{1}{4\sqrt{1+t}}, \frac{1}{4\sqrt{1-t}}, 0 \right), \\ \mathbf{r}'''(t) &= \left(-\frac{1}{8(1+t)^{\frac{3}{2}}}, \frac{1}{8(1-t)^{\frac{3}{2}}}, 0 \right). \end{aligned} \quad (12)$$

由于

$$|\mathbf{r}'(t)|^2 = \frac{1+t}{4} + \frac{1-t}{4} + \frac{1}{2} = 1,$$

所以由定理??知 t 是弧长参数. 因此由曲线论基本公式可知, 所求曲线的曲率、单位切向量、主法向量、副法向量分别为

$$\begin{aligned} \kappa(t) &= |\mathbf{r}''(t)| = \sqrt{\frac{1}{16(1+t)} + \frac{1}{16(1-t)}} = \frac{1}{2\sqrt{2(1-t^2)}}, \\ \boldsymbol{\alpha}(t) &= \mathbf{r}'(t) = \left(\frac{\sqrt{1+t}}{2}, -\frac{\sqrt{1-t}}{2}, \frac{1}{\sqrt{2}} \right), \\ \boldsymbol{\beta}(t) &= \frac{\mathbf{r}''(t)}{|\mathbf{r}''(t)|} = \left(\frac{\sqrt{1-t}}{\sqrt{2}}, \frac{\sqrt{1+t}}{\sqrt{2}}, 0 \right), \\ \boldsymbol{\gamma}(t) &= \boldsymbol{\alpha}(t) \times \boldsymbol{\beta}(t) = \left(-\frac{\sqrt{1+t}}{2}, \frac{\sqrt{1-t}}{2}, \frac{1}{\sqrt{2}} \right). \end{aligned}$$

故所求曲线的 Frenet 标架为 $\{\mathbf{r}(t); \boldsymbol{\alpha}(t), \boldsymbol{\beta}(t), \boldsymbol{\gamma}(t)\}$. 再由(12)式可得

$$\mathbf{r}'(t) \times \mathbf{r}''(t) = \left(-\frac{1}{4\sqrt{2(1-t)}}, \frac{1}{4\sqrt{2(1+t)}}, \frac{1}{4\sqrt{1-t^2}} \right),$$

$$|\mathbf{r}'(t) \times \mathbf{r}''(t)| = \left| \left(-\frac{1}{4\sqrt{2}(1-t)}, \frac{1}{4\sqrt{2}(1+t)}, \frac{1}{4\sqrt{1-t^2}} \right) \right| = \frac{1}{2\sqrt{2}(1-t^2)},$$

$$(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) = (\mathbf{r}'(t) \times \mathbf{r}''(t)) \cdot \mathbf{r}'''(t) = \frac{1}{16\sqrt{2}(1-t^2)^{\frac{3}{2}}}.$$

于是利用曲线挠率的计算公式可得, 曲线的挠率为

$$\tau(t) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = \frac{\frac{1}{16\sqrt{2}(1-t^2)^{\frac{3}{2}}}}{\frac{1}{8(1-t^2)}} = \frac{1}{2\sqrt{2}(1-t^2)}.$$

□

例题 0.7 设正则参数曲线 $\mathbf{r} = \mathbf{r}(t)$ 的 Frenet 标架是 $\{\mathbf{r}(t); \boldsymbol{\alpha}(t), \boldsymbol{\beta}(t), \boldsymbol{\gamma}(t)\}$, 证明:

$$(\boldsymbol{\alpha}(t), \boldsymbol{\alpha}'(t), \boldsymbol{\alpha}''(t)) \cdot (\boldsymbol{\gamma}(t), \boldsymbol{\gamma}'(t), \boldsymbol{\gamma}''(t)) = \varepsilon \cdot |\boldsymbol{\alpha}'(t)|^3 \cdot |\boldsymbol{\gamma}'(t)|^3,$$

其中 $\varepsilon = \text{sgn}(\tau(t))$.

证明 由求导的链式法则和 Frenet 公式可得

$$\begin{aligned} \boldsymbol{\alpha}'(t) &= \frac{d\boldsymbol{\alpha}(t)}{ds} \cdot \frac{ds}{dt} = \kappa(t) |\mathbf{r}'(t)| \boldsymbol{\beta}(t), \quad \boldsymbol{\gamma}'(t) = \frac{d\boldsymbol{\gamma}(t)}{ds} \cdot \frac{ds}{dt} = -\tau(t) |\mathbf{r}'(t)| \boldsymbol{\beta}(t), \\ \boldsymbol{\alpha}''(t) &= \kappa'(t) |\mathbf{r}'(t)| \boldsymbol{\beta}(t) + \kappa(t) \frac{d|\mathbf{r}'(t)|}{dt} \boldsymbol{\beta}(t) + \kappa(t) |\mathbf{r}'(t)| \boldsymbol{\beta}'(t) \\ &= \kappa'(t) |\mathbf{r}'(t)| \boldsymbol{\beta}(t) + \kappa(t) \frac{d|\mathbf{r}'(t)|}{dt} \boldsymbol{\beta}(t) + \kappa(t) |\mathbf{r}'(t)| \frac{d\boldsymbol{\beta}(t)}{ds} \cdot \frac{ds}{dt} \\ &= \left[\kappa'(t) |\mathbf{r}'(t)| + \kappa(t) \frac{d|\mathbf{r}'(t)|}{dt} \right] \boldsymbol{\beta}(t) + \kappa(t) |\mathbf{r}'(t)|^2 [-\kappa(t) \boldsymbol{\alpha}(t) + \tau(t) \boldsymbol{\gamma}(t)] \\ &= -\kappa^2(t) |\mathbf{r}'(t)|^2 \boldsymbol{\alpha}(t) + \left[\kappa'(t) |\mathbf{r}'(t)| + \kappa(t) \frac{d|\mathbf{r}'(t)|}{dt} \right] \boldsymbol{\beta}(t) + \kappa(t) \tau(t) |\mathbf{r}'(t)|^2 \boldsymbol{\gamma}(t); \\ \boldsymbol{\gamma}''(t) &= \left[-\tau'(t) |\mathbf{r}'(t)| - \tau(t) \frac{d|\mathbf{r}'(t)|}{dt} \right] \boldsymbol{\beta}(t) - \tau(t) |\mathbf{r}'(t)| \boldsymbol{\beta}'(t) \\ &= \left[-\tau'(t) |\mathbf{r}'(t)| - \tau(t) \frac{d|\mathbf{r}'(t)|}{dt} \right] \boldsymbol{\beta}(t) - \tau(t) |\mathbf{r}'(t)| \frac{d\boldsymbol{\beta}(t)}{ds} \cdot \frac{ds}{dt} \\ &= \left[-\tau'(t) |\mathbf{r}'(t)| - \tau(t) \frac{d|\mathbf{r}'(t)|}{dt} \right] \boldsymbol{\beta}(t) - \tau(t) |\mathbf{r}'(t)|^2 [-\kappa(t) \boldsymbol{\alpha}(t) + \tau(t) \boldsymbol{\gamma}(t)] \\ &= \kappa(t) \tau(t) |\mathbf{r}'(t)|^2 \boldsymbol{\alpha}(t) + \left[-\tau'(t) |\mathbf{r}'(t)| - \tau(t) \frac{d|\mathbf{r}'(t)|}{dt} \right] \boldsymbol{\beta}(t) - \tau^2(t) |\mathbf{r}'(t)|^2 \boldsymbol{\gamma}(t). \end{aligned}$$

于是

$$\begin{aligned} (\boldsymbol{\alpha}(t), \boldsymbol{\alpha}'(t), \boldsymbol{\alpha}''(t)) &= \kappa^2(t) \tau(t) |\mathbf{r}'(t)|^3 (\boldsymbol{\alpha}(t), \boldsymbol{\beta}(t), \boldsymbol{\gamma}(t)) = \kappa^2(t) \tau(t) |\mathbf{r}'(t)|^3, \\ (\boldsymbol{\gamma}(t), \boldsymbol{\gamma}'(t), \boldsymbol{\gamma}''(t)) &= -\kappa(t) \tau^2(t) |\mathbf{r}'(t)|^3 (\boldsymbol{\gamma}(t), \boldsymbol{\beta}(t), \boldsymbol{\alpha}(t)) = -\kappa(t) \tau^2(t) |\mathbf{r}'(t)|^3, \\ |\boldsymbol{\alpha}'(t)|^3 \cdot |\boldsymbol{\gamma}'(t)|^3 &= |\kappa(t) |\mathbf{r}'(t)| \boldsymbol{\beta}(t)|^3 \cdot |-\tau(t) |\mathbf{r}'(t)| \boldsymbol{\beta}(t)|^3 = \kappa^3(t) \tau^3(t) |\mathbf{r}'(t)|^6. \end{aligned}$$

故

$$(\boldsymbol{\alpha}(t), \boldsymbol{\alpha}'(t), \boldsymbol{\alpha}''(t)) \cdot (\boldsymbol{\gamma}(t), \boldsymbol{\gamma}'(t), \boldsymbol{\gamma}''(t)) = -\kappa^3(t) \tau^3(t) |\mathbf{r}'(t)|^6 = \text{sgn}(\tau(t)) \cdot |\boldsymbol{\alpha}'(t)|^3 \cdot |\boldsymbol{\gamma}'(t)|^3.$$

□

例题 0.8 设 $\mathbf{r} = \mathbf{r}(s)$ 是以 s 为弧长参数的正则参数曲线, 其挠率 $\tau(s) > 0$. 作曲线 $\tilde{C}$:

$$\tilde{\mathbf{r}}(s) = \int_{s_0}^s \boldsymbol{\gamma}(t) dt,$$

证明: $\tilde{\kappa}(s) = \tau(s)$, $\tilde{\tau}(s) = \kappa(s)$.

证明 由题设可知

$$\tilde{\mathbf{r}}'(s) = \boldsymbol{\gamma}(s), \quad |\tilde{\mathbf{r}}'(s)| = |\boldsymbol{\gamma}(s)| = 1.$$

故由定理??知 s 也是曲线 $\tilde{C}$ 的弧长参数. 再由条件和 Frenet 公式可得

$$\tilde{\mathbf{r}}'(s) = \boldsymbol{\gamma}(s), \quad \tilde{\mathbf{r}}''(s) = \boldsymbol{\gamma}'(s) = -\tau(s)\boldsymbol{\beta}(s),$$

$$\tilde{\mathbf{r}}'''(s) = -\tau'(s)\boldsymbol{\beta}(s) - \tau(s)\boldsymbol{\beta}'(s) = \kappa(s)\tau(s)\boldsymbol{\alpha}(s) - \tau'(s)\boldsymbol{\beta}(s) - \tau^2(s)\boldsymbol{\gamma}(s).$$

进而

$$(\tilde{\mathbf{r}}'(s), \tilde{\mathbf{r}}''(s), \tilde{\mathbf{r}}'''(s)) = -\tau(s)(\boldsymbol{\gamma}(s) \times \boldsymbol{\beta}(s)) \cdot \tilde{\mathbf{r}}'''(s) = -\kappa(s)\tau^2(s)(\boldsymbol{\gamma}(s), \boldsymbol{\beta}(s), \boldsymbol{\alpha}(s)) = \kappa(s)\tau^2(s).$$

于是由曲线论基本公式和曲线挠率的计算公式可得

$$\tilde{\kappa}(s) = |\tilde{\mathbf{r}}''(s)| = |-\tau(s)\boldsymbol{\beta}(s)| = \tau(s), \quad \tilde{\tau}(s) = \frac{(\tilde{\mathbf{r}}'(s), \tilde{\mathbf{r}}''(s), \tilde{\mathbf{r}}'''(s))}{|\tilde{\mathbf{r}}''(s)|^2} = \frac{\kappa(s)\tau^2(s)}{\tau^2(s)} = \kappa(s).$$

□

例题 0.9 设 $\mathbf{r}(t)$ 是单位球面上经度为 t 、纬度为 $\frac{\pi}{2} - t$ 的点的轨迹, 求它的参数方程, 并计算它的曲率和挠率.

解 由题设可知曲线 $\mathbf{r} = \mathbf{r}(t)$ 的参数方程为

$$\mathbf{r}(t) = (\sin t \cos t, \sin^2 t, \cos t).$$

从而

$$\mathbf{r}'(t) = (\cos 2t, \sin 2t, -\sin t), \quad \mathbf{r}''(t) = (-2 \sin 2t, 2 \cos 2t, -\cos t), \quad \mathbf{r}'''(t) = (-4 \cos 2t, -4 \sin 2t, \sin t),$$

$$|\mathbf{r}'(t)| = \sqrt{1 + \sin^2 t}, \quad \mathbf{r}'(t) \times \mathbf{r}''(t) = (-2 \sin^3 t, \cos t(1 + 2 \sin^2 t), 2),$$

$$|\mathbf{r}'(t) \times \mathbf{r}''(t)| = \sqrt{5 + 3 \sin^2 t}, \quad (\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t)) = -6 \sin t.$$

于是由曲线论基本公式和曲线挠率的计算公式可得

$$\kappa(s) = \frac{|\mathbf{r}'(t) \times \mathbf{r}''(t)|}{|\mathbf{r}'(t)|^3} = \frac{\sqrt{5 + 3 \sin^2 t}}{(1 + \sin^2 t)^{\frac{3}{2}}}, \quad \tau(s) = \frac{(\mathbf{r}'(t), \mathbf{r}''(t), \mathbf{r}'''(t))}{|\mathbf{r}'(t) \times \mathbf{r}''(t)|^2} = -\frac{6 \sin t}{5 + 3 \sin^2 t}.$$

□

例题 0.10 设 $\{\mathbf{r}(t); \mathbf{e}_1(t), \mathbf{e}_2(t), \mathbf{e}_3(t)\}$ 是沿曲线 $\mathbf{r}(t)$ 定义的一个单位正交标架场, 假定

$$\mathbf{e}_i'(t) = \sum_{j=1}^3 \lambda_{ij}(t) \mathbf{e}_j(t), \quad 1 \leq i \leq 3.$$

证明: $\lambda_{ij}(t) + \lambda_{ji}(t) = 0, i, j = 1, 2, 3$.

证明 根据条件, 再利用求导的链式法则和 Frenet 公式可得

$$\mathbf{e}_1'(t) = \frac{d\mathbf{e}_1(t)}{ds} \cdot \frac{ds}{dt} = |\mathbf{r}'(t)| \kappa(t) \mathbf{e}_2(t) = \sum_{j=1}^3 \lambda_{1j}(t) \mathbf{e}_j(t),$$

$$\mathbf{e}_2'(t) = \frac{d\mathbf{e}_2(t)}{ds} \cdot \frac{ds}{dt} = -|\mathbf{r}'(t)| \kappa(t) \mathbf{e}_1(t) + |\mathbf{r}'(t)| \tau(t) \mathbf{e}_3(t) = \sum_{j=1}^3 \lambda_{2j}(t) \mathbf{e}_j(t),$$

$$\mathbf{e}_3'(t) = \frac{d\mathbf{e}_3(t)}{ds} \cdot \frac{ds}{dt} = -|\mathbf{r}'(t)| \tau(t) \mathbf{e}_2(t) = \sum_{j=1}^3 \lambda_{3j}(t) \mathbf{e}_j(t).$$

由 $\mathbf{e}_1(t), \mathbf{e}_2(t), \mathbf{e}_3(t)$ 的单位正交性可得

$$\lambda_{12}(t) = |\mathbf{r}'(t)| \kappa(t), \quad \lambda_{13}(t) = 0,$$

$$\lambda_{21}(t) = -|\mathbf{r}'(t)| \kappa(t), \quad \lambda_{23}(t) = |\mathbf{r}'(t)| \tau(t),$$

$$\lambda_{31}(t) = 0, \quad \lambda_{32}(t) = -|\mathbf{r}'(t)| \tau(t).$$

故

$$\lambda_{ij} + \lambda_{ji} = 0, \quad i, j = 1, 2, 3.$$

□